

Performance benchmarks using the Rust Scientific Library (Russell)

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We test the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where $\mathbf{A}$ is the coefficient matrix, $\mathbf{x}$ is the solution vector, and $\mathbf{b}$ is the right-hand side vector. The coefficient matrix is set with matrices from the [SuiteSparse Matrix Collection](#). The right-hand side vector is filled with ones, i.e., we study the solution of

$$\mathbf{A}\mathbf{x} = \mathbf{1} \quad (1)$$

The relative error is calculated as

$$\text{RelativeError} = \frac{\max(|\mathbf{A}\mathbf{x} - \mathbf{1}|)}{\max(|\mathbf{A}| + 1)} \quad (2)$$

The tested real-valued matrices are:

- 1 **bwm2000** (Bai) – Brusselator wave model in transport interaction of chemical solutions (1992)
- 2 **rdb5000** (Bai) – Reaction-diffusion brusselator model (1994)
- 3 **Goodwin_040** (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
- 4 **fp** (MKS) – 2-D Fokker Planck eqn, electron dyn. in external field (2006)
- 5 **xenon1** (Ronis) – Complex zeolite, sodalite crystals (2001)
- 6 **twotone** (ATandT) – Harmonic balance method (2001)
- 7 **Raj1** (Rajat) – Circuit Simulation Problem (2007)
- 8 **boyd2** (GHS_indef) – Optimization problem (2004)
- 9 **Goodwin_071** (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
- 10 **darcy003** (GHS_indef) – Discretization using mixed FE of Darcy (2002)
- 11 **rma10** (Bova) – 3D CFD model, Charleston harbor (1997)
- 12 **helm2d03** (GHS_indef) – Helmholtz eq on a unit square (2004)
- 13 **stomach** (Norris) – 3D electro-physical model of a duodenum (2003)
- 14 **oilpan** (GHS_psdef) – Structural problem (2004)
- 15 **ASIC_680k** (Sandia) – Circuit simulation matrix (2006)
- 16 **tmt_unsym** (CEMW) – Electromagnetics problem (2008)
- 17 **Goodwin_127** (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
- 18 **pre2** (ATandT) – Harmonic balance method (2001)
- 19 **marine1** (Martin) – Chemical oceanography; a marine nitrogen cycle inverse model (2018)
- 20 **torso1** (Norris) – Finite differences and boundary element, 2D model of torso (2003)
- 21 **atmosmodd** (Bourchtein) – CFD analysis of atmospheric models (2009)
- 22 **atmosmodl** (Bourchtein) – CFD analysis of atmospheric models (2009)
- 23 **memchip** (Freescale) – Circuit simulation problem (2010)
- 24 **Freescale1** (Freescale) – Circuit simulation problem (2008)
- 25 **rajat31** (Rajat) – Circuit simulation problem (2006)
- 26 **Transport** (Janna) – 3D finite element flow and transport (2012)
- 27 **inline_1** (GHS_psdef) – Structural problem, stiffness matrix (2004)
- 28 **PFlow_742** (Janna) – 3D pressure-temperature evolution in porous media (2014)
- 29 **Emilia_923** (Janna) – Geomechanical model for CO2 sequestration (2011)
- 30 **dielFilterV2real** (Dziekonski) – FEM in electromagnetics (2011)
- 31 **Flan_1565** (Janna) – Structural problem, 3D model of a steel flange (2011)
- 32 **pres-cylin** (Pedroso) – FEM stiffness matrix of a pressurized cylinder (Tet10 with 1,711,464 DOF). Not from the Collection. From [Pedroso DM \(2024\) Caveats of three direct linear solvers for finite element analyses](#).

The tested complex-valued matrices are:

- 1 **mhd1280b** (Bai) – Alfven spectra in magnetohydrodynamics (1994)

- 2 mplate (Cote) – Vibro-acoustic problem (1997)
- 3 RFdevice (Rost) – Semiconductor device simulation (2007)
- 4 vfem (CEMW) – Electromagnetics; vector finite element (2008)
- 5 fem_filter (Lee) – FEM band-pass microwave filter 500MHz (2008)
- 6 Chevron4 (Chevron) – Temporal freq domain seismic modeling (2012)
- 7 mono_500Hz (FreeFieldTechnologies) – 3D vibro-acoustic problem, aircraft engine nacelle (2008)
- 8 kim2 (Kim) – 2D 676-by-676 complex mesh (2002)
- 9 fem_hifreq_circuit (Lee) – FEM Maxwell equations for hi-freq circuit (2009)
- 10 dielFilterV3clx (Dziekonski) – High-order vector finite element method in EM (2011)

Additional notes:

- Each problem is solved ten times. The reported error is the maximum among runs. The reported computer time is the average without outliers.
- Total time includes initialization (memory allocation + symbolic factorization), numeric factorization, and solve.
- Column NNZ (number of non-zeros) corresponds to Pattern Entries reported by the [SuiteSparse Matrix Collection](#).
- Column Sym shows if symmetry information was provided to the solver. An asterisk means positive definite information was also given.
- oom (out-of-memory) indicates that the symbolic factorization was terminated due to insufficient memory.
- Bold values highlight significant results, such as large errors or high computation times.
- The hybrid memory mode is enabled for the pres-cylin matrix and cuDSS.

Information about the system:

```

--- OS ---
NAME="Ubuntu"
VERSION_ID="24.04"
KERNEL=6.17.0-35-generic

--- GPU ---
GPU[0]: NVIDIA GeForce RTX 4090, 595.71.05, 24564 MiB

--- CPU ---
Architecture      : x86_64
CPU(s)            : 32
On-line CPU(s) list: 0-31
Model name        : 13th Gen Intel(R) Core(TM) i9-13900KF
Thread(s) per core : 2
Core(s) per socket : 24
Socket(s)         : 1
CPU(s) scaling MHz : 29%
CPU max MHz       : 5800.0000
CPU min MHz       : 800.0000
BogoMIPS          : 5990.40
L1d cache         : 896 KiB (24 instances)
L1i cache         : 1.3 MiB (24 instances)
L2 cache          : 32 MiB (12 instances)
L3 cache          : 36 MiB (1 instance)
NUMA node0 CPU(s) : 0-31
Vulnerability L1tf : Not affected

--- Memory ---
MemTotal:        32621948 kB
MemFree:         7862028 kB
MemAvailable:    26775568 kB
SwapTotal:       33554428 kB

```

Information about the libraries:

```

cuDSS: 0.8.0.10 (CUDA 13)
MUMPS: 5.9.0
SuiteSparse: latest (from GitHub)

```

The results are given in Tables [1](#), [2](#), [3](#), and [4](#).

Table 1: Calculations on Ubuntu. cuDSS and MUMPS. Real-valued matrices.

Matrix	Nrow	NNZ	Sym	cuDSS			MUMPS	
				Matching Algorithm	Total Time	Relative Error	Total Time	Relative Error
bwm2000	2,000	7,996	No	Auto	8.459ms	8.51×10^{-16}	4.508ms	7.50×10^{-16}
rdb5000	5,000	29,600	No	Auto	30.294ms	2.26×10^{-13}	11.966ms	5.61×10^{-13}
Goodwin_040	17,922	561,677	No	MaxMinDiag	72.731ms	2.88×10^{-12}	97.666ms	1.04×10^{-11}
fp	7,548	848,553	No	Auto	164.293ms	3.31×10^{-9}	135.436ms	1.03×10^{-20}
xenon1	48,600	1,181,120	No	Auto	221.279ms	1.27×10^{-39}	241.041ms	7.13×10^{-40}
twotone	120,750	1,224,224	No	Auto	803.238ms	2.70×10^{-13}	833.388ms	2.11×10^{-14}
Raj1	263,743	1,302,464	No	Auto	957.424ms	1.16×10^{-11}	846.641ms	3.27×10^{-13}
boyd2	466,316	1,500,397	Yes	None	1.111s	3.67×10^{-11}	870.439ms	4.15×10^{-11}
Goodwin_071	56,021	1,797,934	No	MaxMinDiag	141.818ms	1.16×10^{-11}	257.613ms	4.97×10^{-11}
darcy003	389,874	2,101,242	Yes	MaxMinDiagAlt	2.633s	7.32×10^{-7}	10.756s	1.88×10^{-10}
rma10	46,835	2,374,001	No	Auto	252.016ms	6.04×10^{-16}	165.467ms	1.04×10^{-16}
helm2d03	392,257	2,741,935	Yes	None	1.315s	1.26×10^{-10}	1.381s	2.04×10^{-10}
stomach	213,360	3,021,648	No	Auto	1.368s	2.61×10^{-15}	1.467s	1.21×10^{-15}
oilpan	73,752	3,597,188	Yes*	None	107.239ms	4.47×10^{-15}	162.017ms	2.31×10^{-15}
ASIC_680k	682,862	3,871,773	No	Auto	3.959s	7.33×10^{-11}	1m24.044s	1.11×10^{-10}
tmt_unsym	917,825	4,584,801	No	Auto	2.631s	3.64×10^{-7}	2.843s	1.31×10^{-7}
Goodwin_127	178,437	5,778,545	No	MaxMinDiag	419.198ms	7.19×10^{-11}	838.568ms	7.88×10^{-10}
pre2	659,033	5,959,282	No	Auto	3.526s	2.15×10^{-14}	4.235s	2.17×10^{-13}
marine1	400,320	6,226,538	No	Auto	4.204s	4.84×10^{-8}	4.864s	8.90×10^{-14}
torso1	116,158	8,516,500	No	MaxDiagCount	1.144s	3.59×10^{-7}	1.728s	6.97×10^{-7}
atmosmodd	1,270,432	8,814,880	No	Auto	19.346s	1.44×10^{-16}	35.662s	1.41×10^{-16}
atmosmodl	1,489,752	10,319,760	No	Auto	18.698s	8.97×10^{-18}	33.015s	8.70×10^{-18}
memchip	2,707,524	14,810,202	No	Auto	7.310s	3.23×10^{-15}	8.004s	2.50×10^{-15}
Freescape1	3,428,755	18,920,347	No	Auto	8.551s	9.39×10^{-10}	10.269s	9.39×10^{-10}
rajat31	4,690,002	20,316,253	No	Auto	18.008s	3.43×10^{-14}	18.330s	8.15×10^{-15}
Transport	1,602,111	23,500,731	No	Auto	21.856s	7.13×10^{-10}	35.495s	4.56×10^{-10}
inline_1	503,712	36,816,342	Yes*	None	1.848s	4.76×10^{-15}	3.274s	3.59×10^{-15}
PFlow_742	742,793	37,138,461	Yes*	None	10.094s	1.74×10^{-10}	10.843s	4.14×10^{-11}
Emilia_923	923,136	41,005,206	Yes*	None	15.960s	1.54×10^{-22}	32.374s	4.02×10^{-23}
dielFilterV2	1,157,456	48,538,952	Yes	None	7.164s	1.42×10^{-11}	12.179s	1.17×10^{-11}
Flan_1565	1,564,794	117,406,044	Yes*	None	10.383s	2.42×10^{-15}	19.035s	1.29×10^{-15}
pres-cylin	1,711,464	133,562,188	Yes*	None	1m59.325s	9.02×10^{-13}	1m5.998s	7.39×10^{-13}

Table 2: Calculations on Ubuntu. cuDSS and MUMPS. Complex-valued matrices.

Matrix	Nrow	NNZ	Sym	cuDSS			MUMPS	
				Matching Algorithm	Total Time	Relative Error	Total Time	Relative Error
mhd1280b	1,280	12,029	No	Auto	7.256ms	1.57×10^{-15}	7.037ms	1.17×10^{-15}
mplate	5,962	142,190	Yes	None	79.501ms	1.25×10^{-10}	60.236ms	9.34×10^{-11}
RFdevice	74,104	365,580	No	MaxDiagSum	589.147ms	2.91×10^{-2}	1.552s	4.35×10^{-8}
vfem	93,476	1,434,636	No	Auto	882.671ms	5.73×10^{-8}	1.222s	1.11×10^{-7}
fem_filter	74,062	1,731,206	No	Auto	762.298ms	1.40×10^{-10}	1.035s	1.69×10^{-10}
Chevron4	711,450	6,376,412	No	Auto	2.596s	8.66×10^{-11}	2.990s	3.89×10^{-11}
mono_500Hz	169,410	5,036,288	No	Auto	2.806s	1.87×10^{-10}	5.001s	5.12×10^{-10}
kim2	456,976	11,330,020	No	Auto	14.384s	0.00×10^0	4.336s	2.96×10^{-18}
fem_hifreq_circuit	491,100	20,239,237	No	Auto	4.856s	1.83×10^{-10}	9.773s	1.59×10^{-10}
dielFilterV3clx	420,408	32,886,208	Yes	None	2.203s	5.75×10^{-10}	3.879s	6.39×10^{-11}

Table 3: Calculations on Ubuntu. UMFPACK and MUMPS. Real-valued matrices.

Matrix	Nrow	NNZ	Sym	UMFPACK		MUMPS	
				Total Time	Relative Error	Total Time	Relative Error
bwm2000	2,000	7,996	No	716.819 μ s	5.30×10^{-16}	4.508ms	7.50×10^{-16}
rdb5000	5,000	29,600	No	11.897ms	1.93×10^{-15}	11.966ms	5.61×10^{-13}
Goodwin_040	17,922	561,677	No	114.864ms	6.62×10^{-13}	97.666ms	1.04×10^{-11}
fp	7,548	848,553	No	165.520ms	2.57×10^{-21}	135.436ms	1.03×10^{-20}
xenon1	48,600	1,181,120	No	690.129ms	5.55×10^{-40}	241.041ms	7.13×10^{-40}
twotone	120,750	1,224,224	No	368.835ms	1.13×10^{-14}	833.388ms	2.11×10^{-14}
Raj1	263,743	1,302,464	No	oom	oom	846.641ms	3.27×10^{-13}
boyd2	466,316	1,500,397	Yes	1m19.182s	2.86×10^{-13}	870.439ms	4.15×10^{-11}
Goodwin_071	56,021	1,797,934	No	469.794ms	2.77×10^{-12}	257.613ms	4.97×10^{-11}
darcy003	389,874	2,101,242	Yes	1.175s	6.95×10^{-12}	10.756s	1.88×10^{-10}
rma10	46,835	2,374,001	No	446.181ms	1.57×10^{-17}	165.467ms	1.04×10^{-16}
helm2d03	392,257	2,741,935	Yes	1.522s	2.37×10^{-13}	1.381s	2.04×10^{-10}
stomach	213,360	3,021,648	No	1.887s	5.13×10^{-16}	1.467s	1.21×10^{-15}
oilpan	73,752	3,597,188	Yes*	539.295ms	1.26×10^{-15}	162.017ms	2.31×10^{-15}
ASIC.680k	682,862	3,871,773	No	1.384s	2.20×10^{-11}	1m24.044s	1.11×10^{-10}
tmt_unsym	917,825	4,584,801	No	3.341s	5.96×10^{-8}	2.843s	1.31×10^{-7}
Goodwin_127	178,437	5,778,545	No	1.444s	6.03×10^{-12}	838.568ms	7.88×10^{-10}
pre2	659,033	5,959,282	No	oom	oom	4.235s	2.17×10^{-13}
marine1	400,320	6,226,538	No	oom	oom	4.864s	8.90×10^{-14}
torso1	116,158	8,516,500	No	1.554s	3.82×10^{-8}	1.728s	6.97×10^{-7}
atmosmodd	1,270,432	8,814,880	No	oom	oom	35.662s	1.41×10^{-16}
atmosmodl	1,489,752	10,319,760	No	oom	oom	33.015s	8.70×10^{-18}
memchip	2,707,524	14,810,202	No	18.701s	2.04×10^{-15}	8.004s	2.50×10^{-15}
Freescape1	3,428,755	18,920,347	No	29.202s	7.05×10^{-10}	10.269s	9.39×10^{-10}
rajat31	4,690,002	20,316,253	No	oom	oom	18.330s	8.15×10^{-15}
Transport	1,602,111	23,500,731	No	oom	oom	35.495s	4.56×10^{-10}
inline_1	503,712	36,816,342	Yes*	oom	oom	3.274s	3.59×10^{-15}
PFlow_742	742,793	37,138,461	Yes*	oom	oom	10.843s	4.14×10^{-11}
Emilia_923	923,136	41,005,206	Yes*	oom	oom	32.374s	4.02×10^{-23}
dielFilterV2	1,157,456	48,538,952	Yes	oom	oom	12.179s	1.17×10^{-11}
Flan_1565	1,564,794	117,406,044	Yes*	oom	oom	19.035s	1.29×10^{-15}
pres-cylin	1,711,464	133,562,188	Yes*	oom	oom	1m5.998s	7.39×10^{-13}

Table 4: Calculations on Ubuntu. UMFPACK and MUMPS. Complex-valued matrices.

Matrix	Nrow	NNZ	Sym	UMFPACK		MUMPS	
				Total Time	Relative Error	Total Time	Relative Error
mhd1280b	1,280	12,029	No	406.147 μ s	7.41×10^{-16}	7.037ms	1.17×10^{-15}
mplate	5,962	142,190	Yes	712.412ms	1.86×10^{-11}	60.236ms	9.34×10^{-11}
RFdevice	74,104	365,580	No	4.116s	1.22×10^{-13}	1.552s	4.35×10^{-8}
vfem	93,476	1,434,636	No	3.846s	4.70×10^{-8}	1.222s	1.11×10^{-7}
fem_filter	74,062	1,731,206	No	2.892s	6.61×10^{-11}	1.035s	1.69×10^{-10}
Chevron4	711,450	6,376,412	No	6.818s	1.60×10^{-11}	2.990s	3.89×10^{-11}
mono_500Hz	169,410	5,036,288	No	oom	oom	5.001s	5.12×10^{-10}
kim2	456,976	11,330,020	No	oom	oom	4.336s	2.96×10^{-18}
fem_hifreq_circuit	491,100	20,239,237	No	oom	oom	9.773s	1.59×10^{-10}
dielFilterV3clx	420,408	32,886,208	Yes	oom	oom	3.879s	6.39×10^{-11}